Equivalent Fractions and Mixed Numbers

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS equivalent *(ōrite)* · simplify *(whakamāmā)* · improper fraction *(hautanga tauraro-iti)*

A Equivalent fractions ●○○

Remember: Multiply or divide the top and the bottom by the same number. The value of the fraction does not change.

1. Fill in the box to make the fractions equal.

a) $\frac{2}{3} = \frac{\square}{12}$ **b)** $\frac{3}{5} = \frac{\square}{15}$ **c)** $\frac{1}{4} = \frac{5}{\square}$ **d)** $\frac{5}{6} = \frac{\square}{12}$

e) $\frac{3}{8} = \frac{9}{\square}$ **f)** $\frac{4}{7} = \frac{\square}{35}$ **g)** $\frac{7}{10} = \frac{\square}{100}$ **h)** $\frac{5}{9} = \frac{20}{\square}$

i) $\frac{1}{2} = \frac{\square}{18}$ **j)** $\frac{2}{9} = \frac{12}{\square}$ **k)** $\frac{11}{12} = \frac{\square}{36}$ **l)** $\frac{3}{4} = \frac{\square}{100}$

B Simplifying fractions ●○○

Remember: Divide the top and the bottom by their highest common factor.

2. Write each fraction in its simplest form.

a) $\frac{6}{8}$ = _____ **b)** $\frac{10}{15}$ = _____ **c)** $\frac{12}{18}$ = _____ **d)** $\frac{9}{21}$ = _____

e) $\frac{20}{28}$ = _____ **f)** $\frac{16}{40}$ = _____ **g)** $\frac{27}{45}$ = _____ **h)** $\frac{36}{60}$ = _____

i) $\frac{35}{50}$ = _____ **j)** $\frac{22}{55}$ = _____ **k)** $\frac{42}{56}$ = _____ **l)** $\frac{75}{100}$ = _____

C Improper fractions to mixed numbers ●●○

Remember: Divide the top by the bottom. The remainder becomes the top of the fraction part.

3. Write each improper fraction as a mixed number.

a) $\frac{7}{2}$ = _____ **b)** $\frac{11}{3}$ = _____ **c)** $\frac{17}{5}$ = _____ **d)** $\frac{9}{4}$ = _____

e) $\frac{23}{6}$ = _____ **f)** $\frac{29}{8}$ = _____ **g)** $\frac{31}{10}$ = _____ **h)** $\frac{40}{7}$ = _____

i) $\frac{19}{3}$ = _____ **j)** $\frac{50}{9}$ = _____ **k)** $\frac{47}{12}$ = _____ **l)** $\frac{38}{5}$ = _____

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D Mixed numbers to improper fractions ●●○

Remember: Multiply the whole number by the bottom, then add the top.

4. Write each mixed number as an improper fraction.

a) $2\frac{1}{3}$ = _____ **b)** $1\frac{3}{4}$ = _____ **c)** $3\frac{2}{5}$ = _____ **d)** $4\frac{1}{2}$ = _____

e) $2\frac{5}{6}$ = _____ **f)** $5\frac{3}{8}$ = _____ **g)** $6\frac{2}{3}$ = _____ **h)** $3\frac{7}{10}$ = _____

i) $7\frac{1}{4}$ = _____ **j)** $4\frac{5}{9}$ = _____ **k)** $10\frac{1}{2}$ = _____ **l)** $2\frac{7}{12}$ = _____

E Comparing fractions ●●●

Remember: Write both fractions with the same denominator, then compare the tops.

5. Which fraction is larger?

a) $\frac{2}{3}$ or $\frac{3}{5}$ **b)** $\frac{3}{4}$ or $\frac{5}{8}$ **c)** $\frac{1}{2}$ or $\frac{4}{9}$

d) $\frac{5}{6}$ or $\frac{7}{9}$ **e)** $\frac{3}{10}$ or $\frac{1}{4}$ **f)** $\frac{7}{12}$ or $\frac{5}{8}$

g) $\frac{4}{5}$ or $\frac{7}{9}$ **h)** $\frac{2}{7}$ or $\frac{1}{3}$ **i)** $\frac{11}{15}$ or $\frac{7}{10}$

6. Write each set of fractions in order, smallest first.

a) $\frac{1}{2}, \frac{3}{8}, \frac{3}{4}$ **b)** $\frac{2}{3}, \frac{5}{6}, \frac{7}{12}$ **c)** $\frac{3}{5}, \frac{1}{2}, \frac{7}{10}$

d) $\frac{2}{3}, \frac{3}{4}, \frac{5}{8}$ **e)** $\frac{1}{3}, \frac{2}{9}, \frac{5}{18}$ **f)** $\frac{3}{4}, \frac{4}{5}, \frac{7}{10}$

★ Extension

7. Write $\frac{84}{126}$ in its simplest form.

8. Write a fraction that is greater than $\frac{1}{3}$ and less than $\frac{1}{2}$.

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Equivalent Fractions and Mixed Numbers

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS equivalent *(ōrite)* · simplify *(whakamāmā)* · improper fraction *(hautanga tauraro-iti)*

A Equivalent fractions ●○○

Remember: Multiply or divide the top and the bottom by the same number. The value of the fraction does not change.

1. Fill in the box to make the fractions equal.

a) $\frac{3}{4} = \frac{\square}{12}$ **b)** $\frac{2}{5} = \frac{\square}{20}$ **c)** $\frac{1}{6} = \frac{5}{\square}$ **d)** $\frac{5}{8} = \frac{\square}{16}$

e) $\frac{2}{7} = \frac{6}{\square}$ **f)** $\frac{4}{9} = \frac{\square}{45}$ **g)** $\frac{3}{10} = \frac{\square}{100}$ **h)** $\frac{5}{6} = \frac{20}{\square}$

i) $\frac{1}{2} = \frac{\square}{14}$ **j)** $\frac{4}{9} = \frac{24}{\square}$ **k)** $\frac{7}{12} = \frac{\square}{36}$ **l)** $\frac{1}{4} = \frac{\square}{100}$

B Simplifying fractions ●○○

Remember: Divide the top and the bottom by their highest common factor.

2. Write each fraction in its simplest form.

a) $\frac{4}{6}$ = _____ **b)** $\frac{12}{15}$ = _____ **c)** $\frac{14}{21}$ = _____ **d)** $\frac{8}{20}$ = _____

e) $\frac{18}{30}$ = _____ **f)** $\frac{15}{40}$ = _____ **g)** $\frac{28}{49}$ = _____ **h)** $\frac{48}{72}$ = _____

i) $\frac{45}{60}$ = _____ **j)** $\frac{26}{65}$ = _____ **k)** $\frac{36}{54}$ = _____ **l)** $\frac{60}{100}$ = _____

C Improper fractions to mixed numbers ●●○

Remember: Divide the top by the bottom. The remainder becomes the top of the fraction part.

3. Write each improper fraction as a mixed number.

a) $\frac{9}{2}$ = _____ **b)** $\frac{10}{3}$ = _____ **c)** $\frac{13}{5}$ = _____ **d)** $\frac{15}{4}$ = _____

e) $\frac{25}{6}$ = _____ **f)** $\frac{27}{8}$ = _____ **g)** $\frac{43}{10}$ = _____ **h)** $\frac{37}{7}$ = _____

i) $\frac{22}{3}$ = _____ **j)** $\frac{52}{9}$ = _____ **k)** $\frac{41}{12}$ = _____ **l)** $\frac{33}{5}$ = _____

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D Mixed numbers to improper fractions ●●○

Remember: Multiply the whole number by the bottom, then add the top.

4. Write each mixed number as an improper fraction.

a) $3\frac{1}{2}$ = _____ **b)** $1\frac{2}{3}$ = _____ **c)** $2\frac{3}{5}$ = _____ **d)** $4\frac{3}{4}$ = _____

e) $3\frac{1}{6}$ = _____ **f)** $2\frac{5}{8}$ = _____ **g)** $5\frac{1}{3}$ = _____ **h)** $4\frac{9}{10}$ = _____

i) $6\frac{3}{4}$ = _____ **j)** $3\frac{4}{9}$ = _____ **k)** $9\frac{1}{2}$ = _____ **l)** $2\frac{5}{12}$ = _____

E Comparing fractions ●●●

Remember: Write both fractions with the same denominator, then compare the tops.

5. Which fraction is larger?

a) $\frac{3}{4}$ or $\frac{2}{3}$ **b)** $\frac{2}{5}$ or $\frac{3}{10}$ **c)** $\frac{1}{2}$ or $\frac{5}{9}$

d) $\frac{5}{6}$ or $\frac{3}{4}$ **e)** $\frac{7}{10}$ or $\frac{3}{4}$ **f)** $\frac{5}{12}$ or $\frac{3}{8}$

g) $\frac{3}{5}$ or $\frac{5}{9}$ **h)** $\frac{3}{7}$ or $\frac{1}{2}$ **i)** $\frac{8}{15}$ or $\frac{3}{5}$

6. Write each set of fractions in order, smallest first.

a) $\frac{1}{2}$, $\frac{5}{8}$, $\frac{1}{4}$ **b)** $\frac{1}{3}$, $\frac{5}{6}$, $\frac{5}{12}$ **c)** $\frac{4}{5}$, $\frac{1}{2}$, $\frac{3}{10}$

d) $\frac{1}{3}$, $\frac{1}{4}$, $\frac{3}{8}$ **e)** $\frac{2}{3}$, $\frac{5}{9}$, $\frac{11}{18}$ **f)** $\frac{1}{4}$, $\frac{1}{5}$, $\frac{3}{10}$

★ Extension

7. Write $\frac{72}{120}$ in its simplest form.

8. Write a fraction that is greater than $\frac{1}{4}$ and less than $\frac{1}{3}$.

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